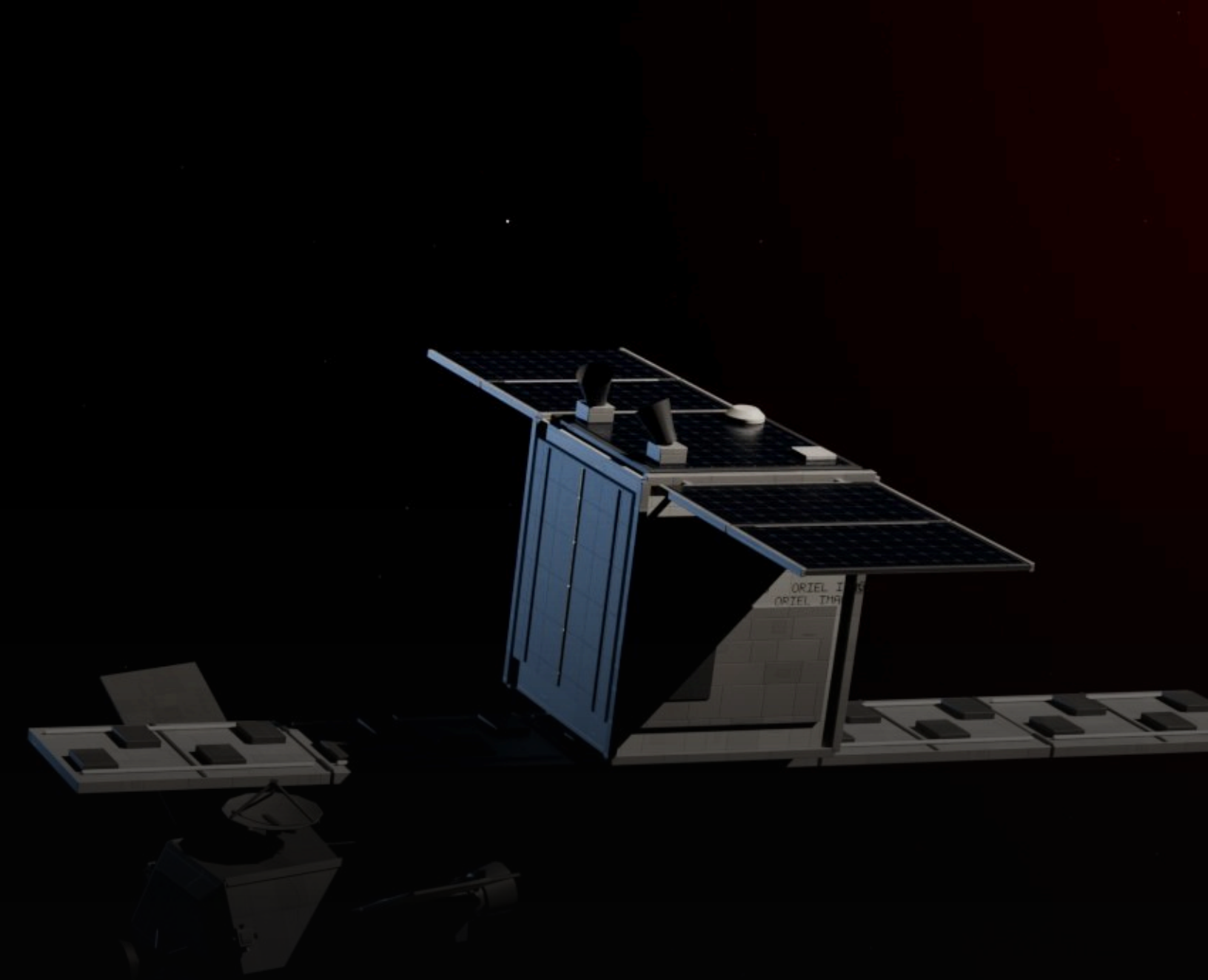


TECHNICAL REPORT · ALPHA 0.5 · 25 OCTOBER 2026

xGEO

RESEARCH EDITION

Physical models, sources and verification in a game of orbital electromagnetic warfare at Jupiter

K. Williamson · xGEO project

BUILD	TESTS	COMPARISONS	ENGINE
Alpha 0.5	63	762 / 762 pass	Godot 4.7

762

of 762 comparisons within tolerance

63

automated tests against independent references

14

physical domains, from orbits to infrared

4

known simplifications, reported, not hidden

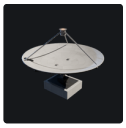
ABSTRACT

xGEO simulates a gray-zone confrontation in low Jupiter orbit in 2041: an uninvited inspector shadows an allied radar satellite, and the player has to make it leave without debris or open war. This edition sets out every physical model the game uses, with its equation, its source, how sure we can be of it, and which numbers are *design values* chosen for play. The game's own code is checked against JPL Horizons, hapsira, scipy, astropy, filterpy, ITU-R recommendations and published worked examples: 762 comparisons across 63 tests, all within tolerance, and 4 further rows that report known simplifications. Section 11 lists what could not be verified.

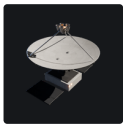
FIG. 1 WHAT THE GAME MODELS, AND WHERE TO READ ABOUT IT



ORBITS & EPHEMERIDES
§2–3 · Horizons, J2, Lambert



RADIO & DETECTION
§4 · ITU patterns, Friis, DF



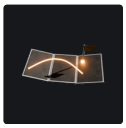
ELECTROMAGNETIC ATTACK
§5 · J/S, spot and barrage



HEAT & SIGNATURES
§6 · energy balance, IR, shadow



TRACKING
§7 · unscented Kalman filter



DOCTRINE & SCENARIO
§8–9 · JP 3-60, JP 3-85

Every model is tested against an independent reference (§10). Click a box to jump to its section.

1 METHOD AND LABELS

The rule is that every number is either sourced or labelled. Each source claim carries one of four labels:

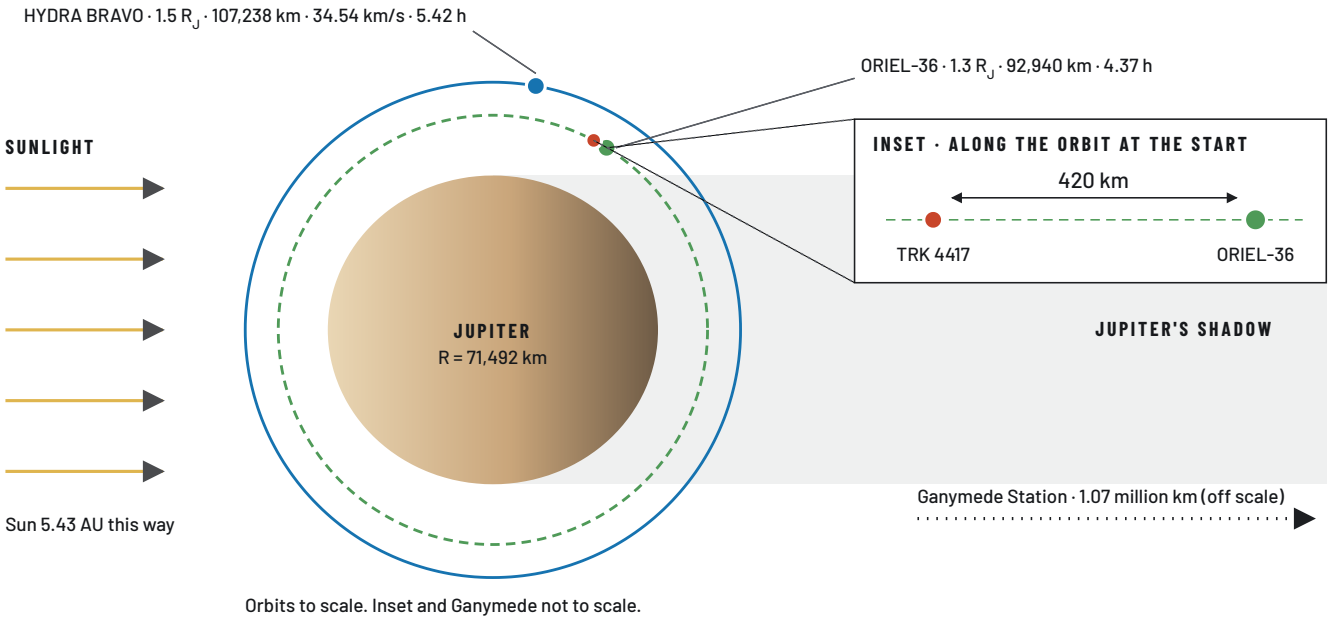
- `V-TEXT` The document was fetched and the passage read; the section, table or page is given.
- `V-RECORD` The work exists (publisher, DOI or official listing) but the passage was not read, because of a paywall or because it is a book. The equation used is the standard textbook form.
- `UNVERIFIED` Not confirmed. Treat it as unsourced.
- `DERIVED` Computed from verified inputs, with the method stated so it can be rerun.

Values chosen for play carry `DESIGN VALUE`. A design value is never presented as measured. Where possible, it is chosen inside a range that a source does support. The most important design values are the J/S needed to break a link (6 dB), the three hours of lost contact after which the inspector withdraws (only with a Blue ship within 25 km of it), the escalation costs, and the compressed timescale of the scenario.

The game runs in Godot 4.7 (GDScript, with 64-bit arithmetic wherever precision matters). Its physics is tested by an external suite (§10), which reads the game's own outputs. Python never reimplements the game in order to grade it.

2 FRAMES, TIME AND EPHEMERIDES

FIG. 2 THE SCENARIO, TO SCALE



HYDRA BRAVO and ORIEL-36 on their circular orbits around oblate Jupiter, with sunlight from the left and the shadow behind. Radii and speeds from §3 and §9; HYDRA's period is DERIVED as 2πr/v.

2.1 Reference frame

The simulation is centred on Jupiter. Its plane is Jupiter's equator, defined by the IAU 2015 pole (WGCCRE[7]), with x toward Jupiter's ascending node on the ICRF equator. The 3D view uses a floating origin at the camera's focus, so positions keep their precision at any distance. The game's clock is UTC. Leap seconds and TT − UTC are ignored, an error of 69.2 s, or 0.16° of Io's orbit.

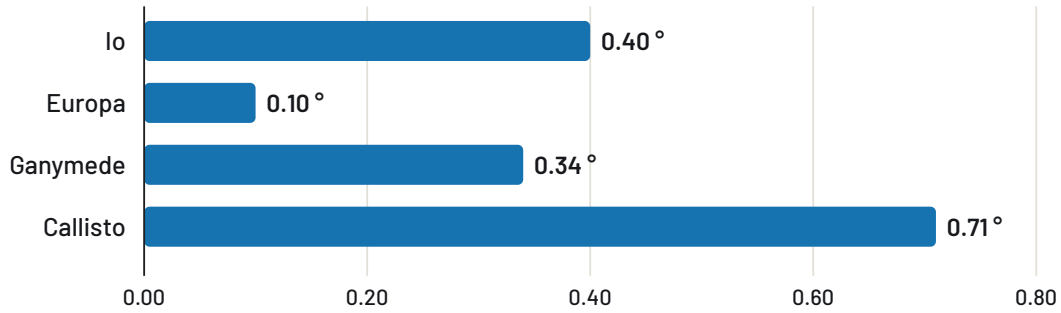
2.2 The Galilean moons

Each moon is a planar Kepler ellipse built from JPL's mean elements (JUP365). The mean longitude advances at the observed sidereal period (NASA's Jovian satellite fact sheet), and the mean anomaly at the period P that JPL tabulates, so periapsis precesses at the difference. JPL's legend calls P the sidereal period; the game uses it as the mean-anomaly period because, paired with the fact-sheet periods, that reproduces Horizons. An earlier build took the mean motion from Kepler's third law, $\sqrt{(\mu/a^3)}$. That ignores J2 and the Laplace resonance, and it put Io 114° from its true place at the campaign start. The current errors against JPL Horizons are:

MOON	PLANAR ERROR, CAMPAIGN	LONGITUDE, CAMPAIGN	LONGITUDE, 2036–2046	OUT OF PLANE (IGNORED)
Io	2,970 km	0.40°	0.43°	340 km
Europa	1,230 km	0.10°	0.24°	5,370 km
Ganymede	6,400 km	0.34°	0.38°	2,490 km
Callisto	23,200 km	0.71°	0.72°	9,900 km

Table 1. Moon positions against JPL Horizons. The Laplace angle $\lambda_I - 3\lambda_E + 2\lambda_G$ follows Horizons to 1.1°.

FIG. 3 **MOON LONGITUDE ERROR AT THE CAMPAIGN**



Against JPL Horizons, from Table 1. Callisto, the farthest, is worst at 0.71°.

2.3 The Sun and planets

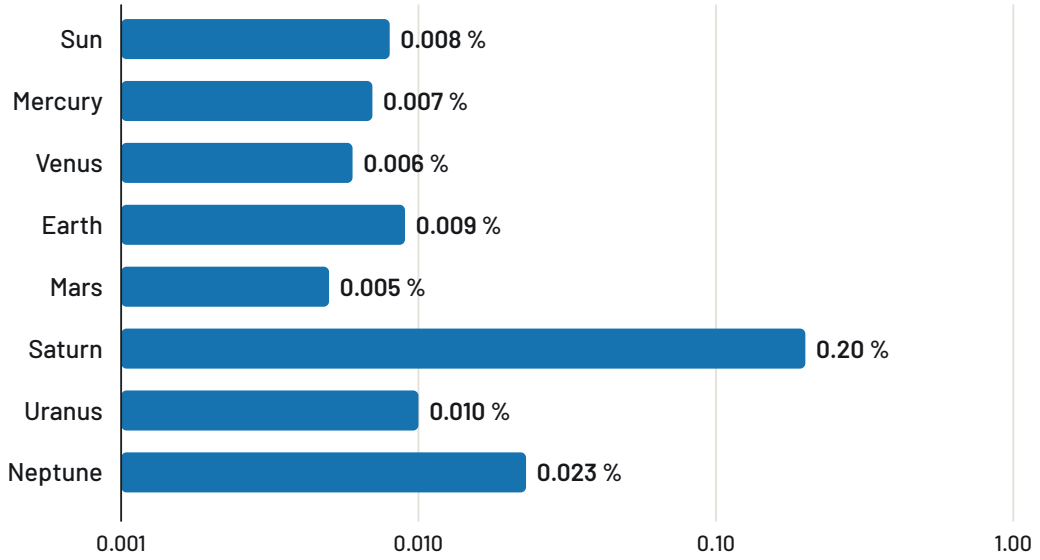
All eight planets come from JPL's approximate Keplerian elements (Standish & Williams[2]): Table 1 for 1800–2050, and Tables 2a and 2b otherwise. They are converted to Jupiter-centred positions in the game frame and cached on a cubic-Hermite grid every 6 h, which adds under 1 km of error. "Earth" in these tables is the Earth–Moon barycentre, 0.0003° from Earth as seen from 5 AU. GM values come from DE440[3]; radii from WGCCRE 2015[7] (Table 4, read through NAIF's pck00011 kernel); and the solar radius is the IAU 2015 nominal 695,700 km[8].

The verification suite checks every planet against Horizons over 2030–2050 and 2060–2199. As a spot check for this edition, the game's distances from Jupiter at the scenario's start (2041-10-12 16:10 UTC) were compared with Horizons vectors (centre 500@599, geometric):

BODY	GAME (AU)	HORIZONS (AU)	DIFFERENCE	ONE-WAY LIGHT TIME
Sun	5.4283	5.4279	0.008%	45.1 min
Mercury	5.7316	5.7312	0.007%	47.7 min
Venus	5.2125	5.2122	0.006%	43.4 min
Earth	6.3711	6.3705	0.009%	53.0 min
Mars	6.6496	6.6493	0.005%	55.3 min
Saturn	4.6212	4.6304	0.20%	38.4 min
Uranus	19.388	19.386	0.010%	2 h 41 min
Neptune	35.232	35.224	0.023%	4 h 53 min

Table 2. Planet distances from Jupiter at the scenario epoch. Saturn's 0.2% (1.4 million km) lies within the stated accuracy of the approximate elements. Saturn is unusually close because Jupiter and Saturn were in conjunction in late 2040.

FIG. 4 PLANET DISTANCES AGAINST HORIZONS



Difference in distance from Jupiter at 2041-10-12 16:10 UTC, from Table 2 (log scale). Every body is within 0.03% except Saturn.

3 GRAVITY AND TRAJECTORY DESIGN

3.1 Force model and integration

The acceleration on a craft is Jupiter's point mass, its J2 zonal term, and the four Galilean moons as third bodies (direct minus indirect term). The Sun is always on. Planets are added beyond 3 million km from Jupiter, where their tidal pull starts to matter. Inside that radius it is under 10^{-7} of Jupiter's (Montenbruck & Gill[20] §3.3, Cowell's method):

$$a = -\mu\, r/|r|^3 + a_{J2}(r) + \Sigma_k\, \mu_k\, [\, (r_k - r)/|r_k - r|^3 - r_k/|r_k|^3\,]$$

Jupiter's GM is 126,687,000 $km^3\ s^{-2}$ (NASA fact sheet[9]; 3.7×10^{-6} from JPL's JUP365). J2 is $14{,}696.514\times10^{-6}$ at R = 71,492 km (Juno, Folkner 2017 Table 2; 3.9×10^{-6} from Iess 2018 Table 1). Integration is RK4 with 60 s substeps near Jupiter. Beyond 3 million km the step adapts to 0.004 × the shortest free-fall time $\sqrt{(d^3/\mu)}$ to any body (Aarseth[21]). The coefficient is a DESIGN VALUE. Against DOP853 at rtol 10^{-13}, the integrator stays within 0.03 m over a day.

3.2 Two-body tools

Kepler propagation uses universal variables (Curtis[17] Alg. 3.3–3.4), with Danby's starter near e = 1 and Vallado's starter for long hyperbolic coasts (Vallado[18] Alg. 8). Lambert's problem is solved by Curtis Alg. 5.2. An earlier loss of precision in the Stumpff function c(z) as $z \to 4\pi^2$ made every transfer over 180° fail. The half-angle form $2\sin^2(\sqrt{z}/2)/z$ fixed it, and all 129 test transfers now agree with Izzo's solver (hapsira) to 1.9×10^{-12}.

3.3 Refining under full gravity

At 1.3 Jupiter radii, J2 bends a 3-hour coast by thousands of kilometres, so a two-body Lambert solution misses its target. Every planned burn is therefore refined by differential-correction shooting through the full-gravity coast (Vallado[18]; single shooting as set out in NASA/TP-20220014814 §5.2.1). Newton's method starts from a two-body Jacobian and updates it with Broyden's method[24]. HYDRA's transfer to ORIEL-36 flies at 3.570 km/s, against 3.869 km/s two-body, and arrives 0.012 km from its aim point. Circular orbits are initialised with $v^2 = \mu/r + 1.5\,\mu\,J2\,R^2/r^3$, so they don't oscillate.

3.4 The planner

The planner shows a porkchop plot: departure wait against time of flight, each cell a refined transfer coloured by the Δv to go and the Δv to match on arrival. For a target that circles in under 12 hours, waits run to 24 h in 1.5 h steps and flights from 1 to 6 h. Those grid steps are design values. The cheapest rendezvous with ORIEL-36 at the start is a wait of 18 h and a flight of 3 h: 1.90 km/s planned, 1.83 km/s flown, then 1.38 km/s to match.

3.5 Interplanetary flight

Where a campaign allows it, the planner plans between planets with patched conics (Curtis ch. 8; Vallado ch. 12). A heliocentric Lambert arc runs between the planets. The escape burn is fitted so its hyperbola leaves from the ship's actual position. Arrival is costed as capture into an orbit at 1.5 planet radii, a design value. The plan is then refined under full gravity, which is necessary because the Sun's tide during escape bends the departure by about 0.3 km/s. Spheres of influence follow Laplace, $r = a(m/M)^{2/5}$ (Curtis §8.4).

The Solar System window shows a minimum-energy (Hohmann) estimate for every body:

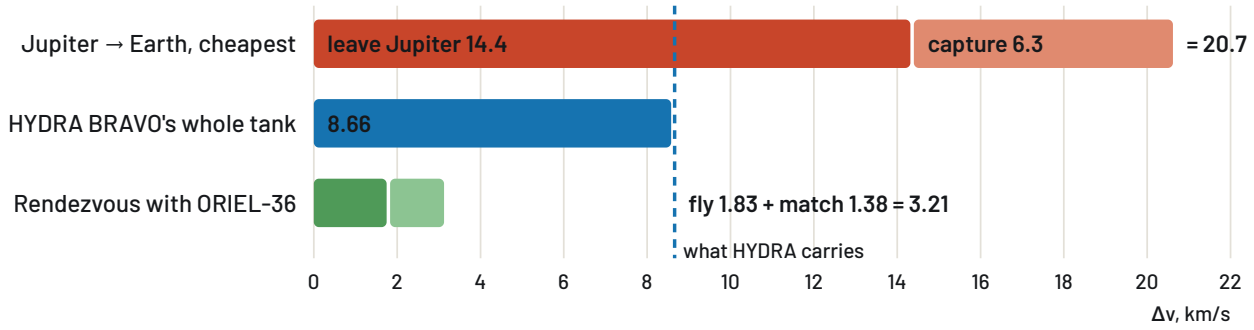
$$t = \pi \sqrt{(a_t^3/\mu_\odot)}, \quad a_t = (r_1 + r_2)/2$$

$$v_\infty = \sqrt{(\mu_\odot/r_1)}\ \left|\sqrt{(2r_2/(r_1+r_2))} - 1\right|$$

$$\Delta v_{leave} = \sqrt{(v_\infty^2 + 2\mu_J/r_s)} - v_s$$ (burn at the ship's radius r_s and speed v_s)

HYDRA orbits at 107,238 km from Jupiter's centre at 34.54 km/s, where the local escape speed is 48.6 km/s. Leaving Jupiter for Earth therefore costs about 14.4 km/s, plus 6.3 km/s to be captured at Earth, against HYDRA's 8.66 km/s. Interplanetary flight is switched off in the first scenario, and these figures show it would be out of reach anyway. That is real physics, not a game limit. From Jupiter's orbit, the cheapest flight takes 3.2 years to Mars and 2.9 years to Earth, and a fall into the Sun takes 2.2 years.

FIG. 5 **WHY HYDRA STAYS AT JUPITER**



Minimum-energy Δv from §3.4–3.5. Leaving Jupiter from HYDRA's orbit costs more than its whole tank; the scenario's rendezvous costs about a third of it.

4 RADIO: ANTENNAS, LINKS AND DETECTION

4.1 Antenna gain and patterns

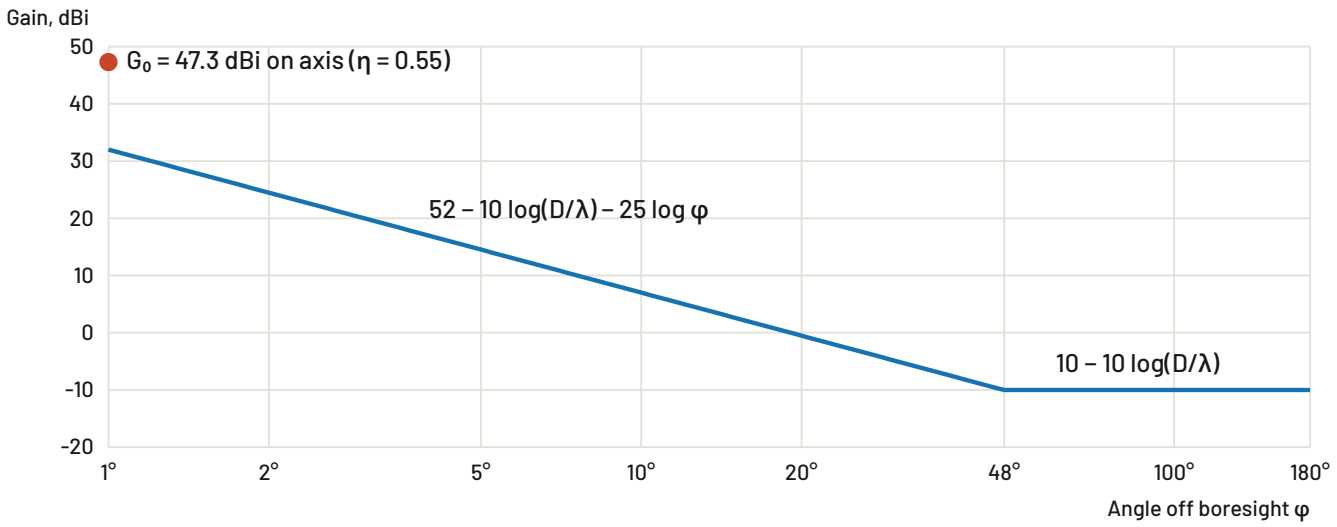
The peak gain of a dish of diameter D at wavelength λ is $G_0 = \eta(\pi D/\lambda)^2$, with $\eta = 0.55$ throughout. That is 1.24 dB conservative against the DSN 34 m antenna at X band (68.24 dBi at 8425 MHz, DSN 810-005 Module 104 Rev. O Table 8[39]). The beamwidth is about 70λ/D degrees (NAWCWD TP 8347 §3-1[54]; ITU-R SA.509-3 §1.3[33]) V-TEXT.

Off-axis gain follows the ITU reference envelopes: SA.509-3 for large dishes ($D/\lambda \geq 100$), and S.465-5 Note 4 for small ones[34]:

```
G(ϕ) = 52 − 10 log(D/λ) − 25 log ϕ   dBi,   100λ/D ≤ ϕ < 48°
G(ϕ) = 10 − 10 log(D/λ)              dBi,   48° ≤ ϕ ≤ 180°
```

These are envelopes of sidelobe peaks, used in interference coordination, so they are conservative for a well-designed antenna. Note 2 of S.465 warns against using them below $D/\lambda = 50$, and every Red dish in the game is that small. They are still the most credible public pattern. Every emitter points: Red's telemetry and command antennas at their handler, civil probes at Earth, and the ore hauler at Ganymede. Warships and buoys listen through four-face arrays that form beams in all directions (after the TDRS S-band multiple-access array, NASA TM-105422). Their gain is the face's peak plus 10 log cos(scan). Ganymede Station's 12 m dish has to be pointed: an assist cues it to the most relevant track, and the player can override it.

FIG. 6 REFERENCE ANTENNA ENVELOPE



ITU-R SA.509-3 sidelobe envelope drawn for an illustrative dish 100 wavelengths across. It is an envelope of sidelobe peaks, so it is conservative for a well-designed antenna.

4.2 Link budget and noise

$$P_r = P_t\, G_t(\theta_e)\, G_r(\theta_r)\, (\lambda / 4\pi R)^2 \qquad \text{(Friis 1946)}$$

$$N = k\, T_s\, B$$

Free-space loss is the exact $20 \log(4\pi R/\lambda)$ of ITU-R P.525-4 eq. (3). The practical eq. (4) rounds its constant to 32.4 dB (exactly 32.448 dB), so the game differs from eq. (4) by 0.048 dB.

4.3 Detection of unknown bursts

Listeners don't know a burst's waveform, so they detect it with an energy (radiometer) detector (Urkowitz 1967[51]). The integrated SNR grows as $\sqrt{(B\tau)}$. The chance of detection is:

$$P_d = Q\left(\left(Q^{-1}(P_{fa}) - d\right) / \sqrt{(1 + 2s)} \right), \qquad P_{fa} = 10^{-6}$$

The false-alarm rate is a **DESIGN VALUE**. This expression matches the exact chi-square distributions to 0.01 for $n \geq 10^5$ samples, and even odds come at 6.8 dB. Spreading a signal over a wider band costs an eavesdropper's energy detector $\sqrt{(B_{old}/B_{new})}$ in SNR at fixed power (the radiometer equation; Urkowitz 1967[51]). ITU-R SM.1055[38] §1 notes that spread signals are noise-like and not readily detected.

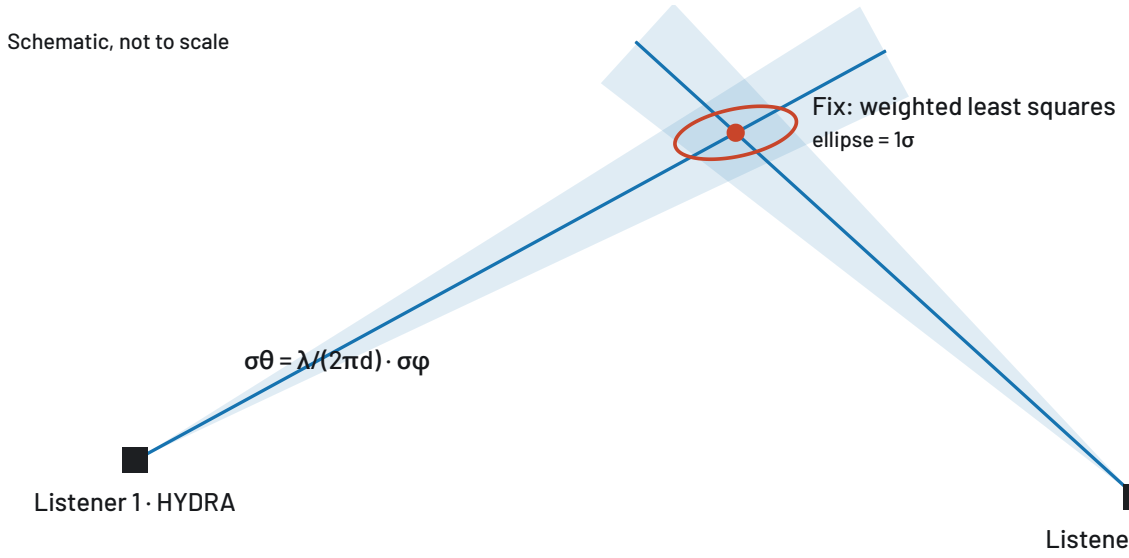
4.4 Direction finding and identification

A two-element interferometer measures a bearing with

$$\sigma_\theta = \lambda/(2\pi d)\cdot\sigma_\phi, \qquad \sigma_\phi^2 = (1+2s)/(2B\tau\, s^2)$$

This is the cross-correlation phase error for a noise-like signal. It follows from the Knapp & Carter (1976) Cramér–Rao bound in terms of coherence, $\gamma^2 = s^2/(1+s)^2$, in the narrowband limit; it is checked by a Monte Carlo of a two-element correlator to 3%. A 20 µrad floor is a DESIGN VALUE. Bearings from separated listeners are crossed by weighted least squares. Identification compares a recording's modulation and symbol rate against a threat library above a minimum SNR, a design-value gate. Specific-emitter identification is real: it recognises a device by imperfections in its analog circuitry (Danev et al.[78]). Once a signal is identified, the coherent turnaround ratio $f_{up} = f_{down} \times 749/880$ (CCSDS 401 Rec. 2.6.2; DSN 810-005[39]) gives its command uplink.

FIG. 7 **DIRECTION FINDING**



Two listeners each take a bearing with its own error wedge. The fix is their weighted least-squares crossing; the farther apart the listeners, the tighter the ellipse. Schematic.

5 ELECTROMAGNETIC ATTACK

5.1 The jamming equation

Jamming a command uplink means being louder, at the victim's receiver, than its handler. Both paths are one-way, so the equation is Friis applied twice. This is the communications form of NAWCWD TP 8347 §4-9 V-TEXT, and

Adamy's EW 101[75] gives it in dB form:

$$J = P_j + G_j(\text{aim error}) + G_v(\theta_j) - L_j$$

$$S = P_u + G_u + G_v(0) - L_s$$

$$J/S = J - S - 10\log(B_j/B_v) \quad \text{dB}$$

$G_v(\theta_j)$ is the victim's gain toward the jammer, at angle θ_j off its line to the handler. From broadside, the victim's discrimination is −44.8 dB. Near the handler's line it falls to −13 dB, so geometry matters. The aim term means a loose track costs beam-pointing gain. HYDRA's jammer is 1 kW from four combined 250 W TWTAs into a 1.5 m dish at 7.18 GHz, 68.5 dBW EIRP. That is a plausible descendant of the Counter Communications System. Real CCS and Meadowlands power figures are not public, so this is an extrapolation DESIGN VALUE.

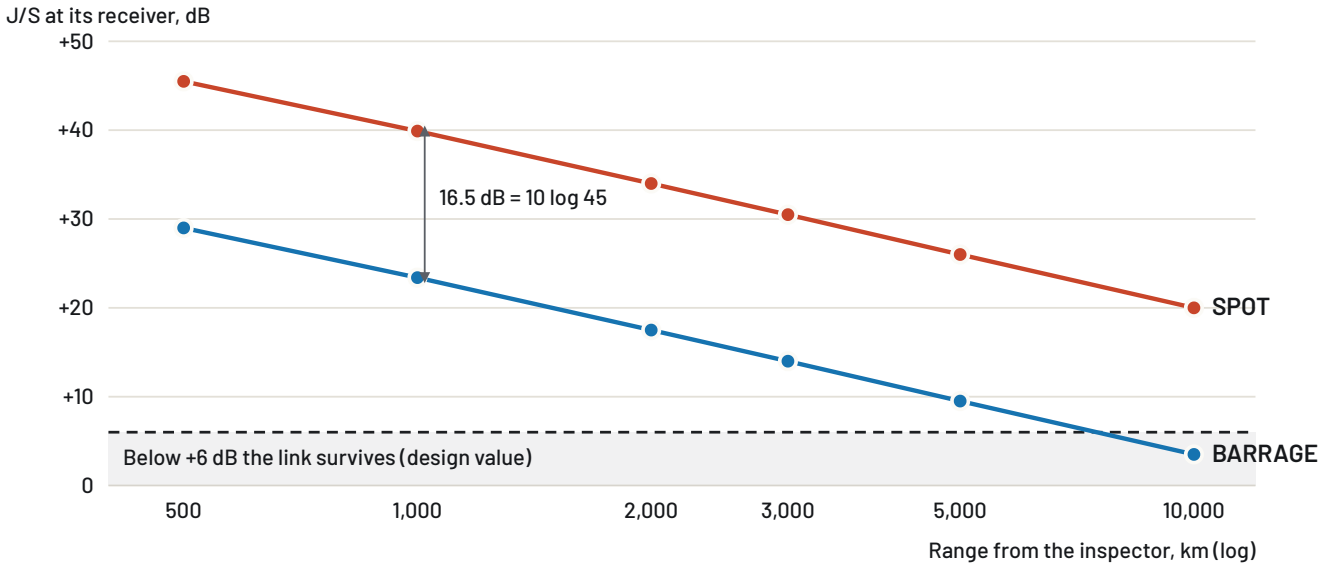
5.2 Spot and barrage

Before the signal is identified, the jammer must spread its power across the whole deep-space X uplink allocation, 7145–7190 MHz (DSN 810-005 Module 201 Rev. F V-TEXT). That costs 10 log(45) = 16.53 dB against a 1 MHz channel. An earlier build used 90 MHz, which included the near-Earth 7190–7235 MHz allocation. Spot jamming of the identified channel removes the penalty. Measured from beside the inspector, barrage gives:

RANGE	500 KM	1,000	2,000	3,000	5,000	10,000
Barrage J/S	+29.0 dB	+23.4	+17.5	+14.0	+9.5	+3.5
Spot J/S	+45.5 dB	+39.9	+34.0	+30.5	+26.0	+20.0

Table 3. J/S against the inspector's command uplink. Spot is barrage + 16.5 dB.

FIG. 8 JAMMING THE INSPECTOR'S COMMAND UPLINK



J/S against range from Table 3. Spot jamming, possible only once the signal is identified, gains 16.5 dB over barrage. The +6 dB threshold is a DESIGN VALUE. Hover a point for its value.

How much J/S breaks a link has no universal number. NAWCWD §4-7 (p. 4-7.3) says plainly that effectiveness "is not a direct mathematical function of J/S" V-TEXT. The game requires +6 dB, a DESIGN VALUE inside the 0–10 dB band typical for FEC-coded BPSK.

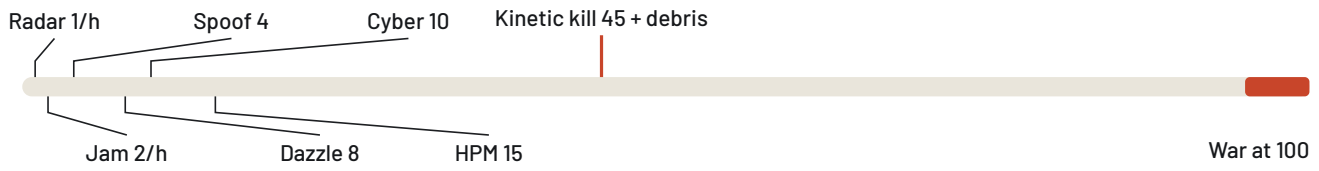
5.3 Other effects

EFFECT	MODEL	SOURCE / STATUS
Spoof (FAKE)	False ranging tones capture the target's ranging loop at J/S ≥ 0 dB, inducing a drift	Loop capture at 0 dB and a drift of 900 km/h are design values
Dazzle	Spot radius $R\sqrt{((1.22\lambda/D)^2 + j^2)}$ at 1.06 µm, 2 m aperture, 0.2 µrad jitter; 10 W/m² at the aperture	Airy disc $1.22\lambda/D$ (Hecht); jitter added in quadrature (engineering approximation); 10 W/m² is an engineering estimate allowing for an OD-6 filter
HPM (PULSE)	Far field $E = \sqrt{(30PG)}/R$ from a 1 GW-peak X-band source; front-door damage at 100 V/m	$E = \sqrt{(30PG)}/R$ (Balanis; ITU-R P.525-4 eq. 1); 100 V/m is a DESIGN VALUE (engineering estimate); GW-class backward-wave oscillators (Benford et al. 2016)
Cyber (TAKE)	Takeover after recording enough command traffic	Abstracted. Real command links are encrypted, and this is stated as fiction
Kinetic kill	$P_k = \text{rel}\cdot(1 - e^{-r^2/2\sigma^2})$, from divert reach against track uncertainty	Rayleigh. Breakup by the NASA standard breakup model (below). The 1,200 catalogued pieces are a DESIGN VALUE set below FY-1C (2007; 3,442 catalogued[138]) and Cosmos 1408 (2021)

A kill breaks the wreck up by the NASA standard breakup model (Johnson et al. 2001[131], as given in Frey & Colombo 2021 App. B[135]). An impact is catastrophic above 40 J/g, and $N(\geq L_c) = 0.1\ M^{0.75}\ L_c^{-1.71}$ fragments are drawn. Each fragment's area-to-mass ratio $\chi = \log_{10}(A/M)$ follows App. B: the bimodal spacecraft law above 11 cm, the small-object law below 8 cm, and a linear blend in $\log L_c$ between 8 and 11 cm. Its speed follows $\log_{10} \Delta v \sim N(0.9\chi + 2.9, 0.4)$, Δv in m/s.

Escalation costs are design values: jamming 2 per hour, radar 1 per hour, spoofing 4, dazzling 8, cyber 10, HPM 15, and a kinetic kill 45. Doctrine provides no escalation ladder (§8).

FIG. 9 THE ESCALATION LADDER



Jamming alone, all day: 2 × 24 = 48 a day. The scenario lasts three days. All costs are DESIGN VALUES.

The costs from §5.3. Doctrine provides no ladder (§8): its ordering, reversible and deniable acts cheap, debris ruinous, is the game's.

6 HEAT, INFRARED AND VISIBLE SIGNATURES

6.1 Energy balance

Each craft has a skin and, if it carries a reactor, radiators. Skin temperature follows a single-node balance with heat capacity (Gilmore[103]):

$$C\,dT/dt = P_{leak} + \alpha\,S\,f\,A/4 - \varepsilon\,\sigma\,T^4\,A, \qquad C = 897\ \mathrm{J\,kg^{-1}\,K^{-1}} \times 0.25 \times m_{dry}$$

S = 50.26 W m^{-2} at Jupiter (NASA fact sheet V-TEXT), and f is the sunlit share (§6.4). The heat capacity is aluminium's at 298 K, 24.21 J mol^{-1} K^{-1} or 897 J kg^{-1} K^{-1} (NIST-JANAF, NIST Chemistry WebBook[105]); Incropera[104] Table A.1 gives 903 J kg^{-1} K^{-1} at 300 K, 0.7% more. The the 25% skin fraction is a DESIGN VALUE. HYDRA's thermal time constant is about 22.5 h, so a pass through Jupiter's shadow cools it only from 190 K to about 187 K. Absorptance and emittance come from NASA RP-1121 (Henninger 1984[95]).

Reactor ships use the 100 kWe Brayton plant of NASA Glenn's JIMO power-system study, at 20.2% efficiency (Mason 2003[96]). That rejects 395 kW, all of it through two-sided Lambertian radiators (JIMO's main radiator carried 364 kWt; the rest left by other routes), $I(\theta) = \varepsilon\sigma T^4 A \cos\theta/\pi$. WARDEN's 85 m^2 per face runs at 462 K, within 1.1% of JIMO's design point (364 kWt from 170 m^2, Siamidis 2005[98]). Radiators lie flat to the plane of play, so every observer in the plane sees only their edge.

6.2 Going dark

A ship can stop radiating for a while by storing its heat. WARDEN evaporates and vents 2 t of water (2.5 MJ/kg, the latent heat near 0 °C), which carries 395 kW for 3.5 h and is then gone. VESPER melts 500 kg of ice (333 kJ/kg, Leimkuehler 2011[102]) for 15.4 h and refreezes it afterwards.

6.3 In-band infrared detection

Telescopes work in NEO Surveyor's NC2 band, 6–10 µm (Mainzer et al. 2023[115] V-TEXT). The in-band share of each body's emission comes from the Widger & Woodall series, which matches astropy to 6×10^{-11}. The 0.5 m LWIR's 1σ noise-equivalent irradiance is 7×10^{-18} W m^{-2}. That lies inside the 0.4–1.1×10^{-17} range derived from NEO Surveyor's S/N-5 sensitivity. A look detects with

$$S/N = I / R^2 / (NEI \times background), \qquad P_d = \Phi(S/N - 5)$$

Against Jupiter's disc, the background raises the NEI ×50, a DESIGN VALUE inside the ×26–92 derived for a background-limited sensor. Within 3 Jupiter radii off the disc, stray light raises it ×5, also a design value. False

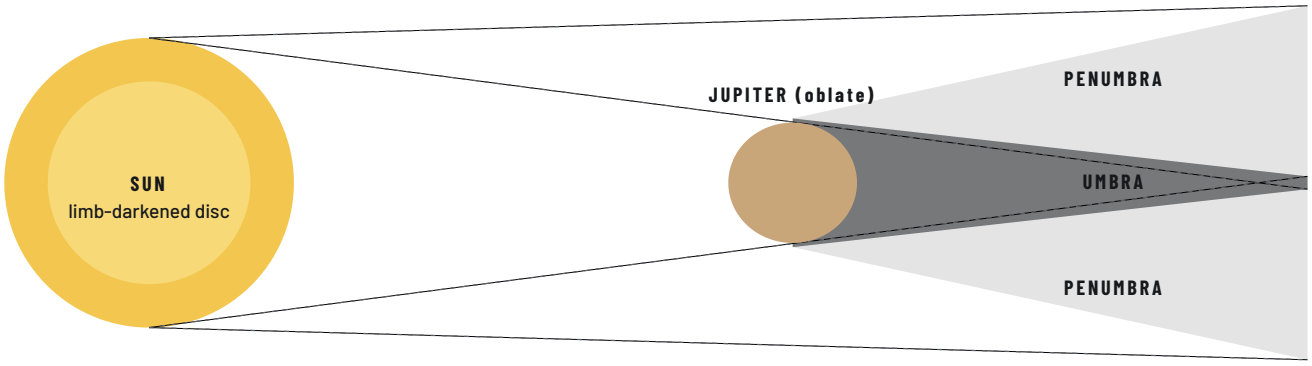
alarms are Poisson, with a mean of 0.1 per sensor per look (design value), and only lines from two places at once start a track.

6.4 Shadows and reflected light

The Sun is a limb-darkened disc, $I(\mu) = 1 - u(1 - \mu)$ with $u = 0.6$ (Allen's[106] §14.7), sampled by 397 rays in 12 rings. The sunlit share is the weighted fraction not blocked by oblate Jupiter (71,492 × 66,854 km), the four moons, or, far out, a planet. This gives both umbra and penumbra (Montenbruck & Gill §3.4.2), within 0.019 of a dense numerical integration.

Sunlit hulls are also seen in visible light. The Sun's $M_V = 4.81$ (Willmer 2018) is scaled to Jupiter's irradiance, and each hull is a Lambert sphere with phase law $\Phi(\alpha) = (\sin\alpha + (\pi - \alpha)\cos\alpha)/\pi$. It returns a share $p \cdot \Phi(\alpha) \cdot r^2/R^2$ of the sunlight at its distance R, with p the geometric albedo and $r^2 = A/4\pi$ for a hull of surface area A (Madhusudhan & Burrows 2012 eq. 3[122]). An earlier build used the cross-section πr^2 in place of r^2, which made every hull π times (1.24 mag) too bright and every visible range $\sqrt{\pi} = 1.77$ times too long. A 1 m GEODSS-class telescope reaches magnitude 22 (Stokes et al. 1996[119]; the free 1998 Lincoln Laboratory Journal account of the same tests gives magnitude 22 at S/N 4 in under 100 s). The game asks for S/N 5 at magnitude 22, a DESIGN VALUE. Visible telescopes look only where a track already exists, because reaching magnitude 22 takes about 100 s staring at one field.

FIG. 10 **UMBRA AND PENUMBRA**



Schematic, not to scale. 397 rays in 12 rings sample the disc; the share not blocked is the sunlit fraction f.

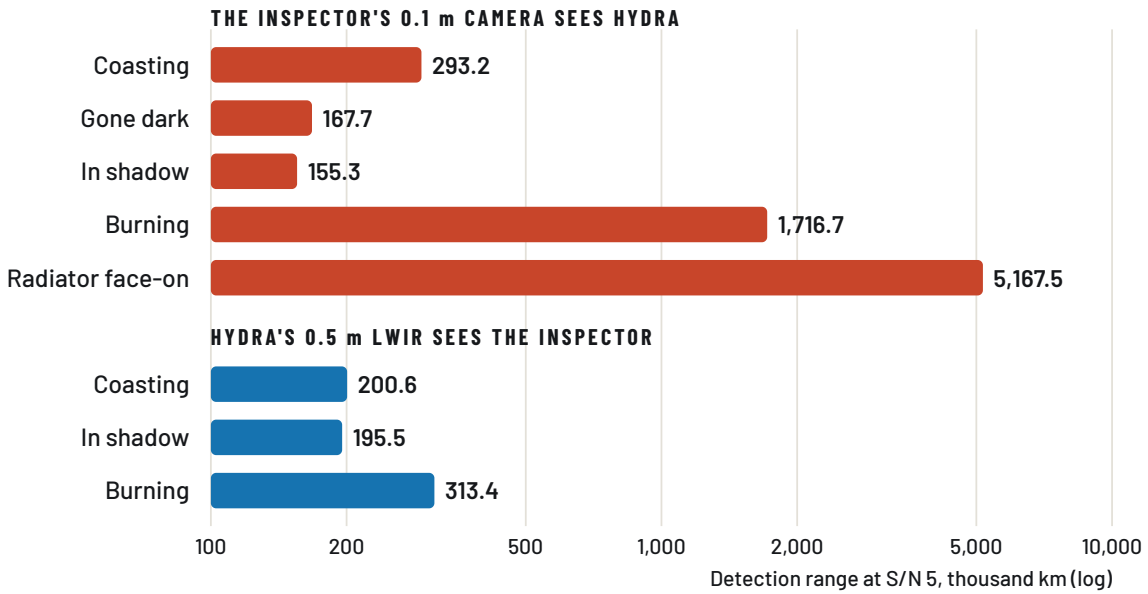
Because the Sun is a disc, Jupiter casts a full shadow inside a soft-edged one (§6.4).

6.5 What the scenario's sensors see

OBSERVER → TARGET	COASTING	DARK	RADIATOR FACE-ON	BURNING	IN SHADOW
HYDRA's 0.5 m LWIR → the inspector	200.6	200.6	200.6	313.4	195.5
The inspector's 0.1 m camera → HYDRA	293.2	167.7	5,167.5	1,716.7	155.3
The inspector's camera → ORIEL-36	181.8				

Table 4. IR detection ranges at S/N 5, thousand km. HYDRA starts out of the inspector's sight, and is seen far sooner coasting than dark.

FIG. 11 **WHO SEES WHOM, AND HOW FAR**



Infrared detection ranges from Table 4 (log scale). Going dark cuts the range at which the inspector sees HYDRA by 43%; a radiator turned toward it makes it about eighteen times longer.

With a 1 m telescope, the visible detection ranges after the corrected Lambert-sphere formula (§6.4) are:

HULL	VISIBLE RANGE, 1 M (THOUSAND KM)
GADFLY	384
CROW	470
WREN ES/DF buoy	271
OWP-62 WARDEN	1,213
PFS	2,093
HMH-5 heavy hauler	3,309
GMC-2 ore carrier	632
LANTERN relay	509
OX ore tug	770
PROSPECTOR probe	272

Table 4b. Visible detection ranges with a 1 m aperture, thousand km, with the $r^2 = A/4\pi$ correction (each is 1/1.77 of the earlier build's).

7 TRACKING

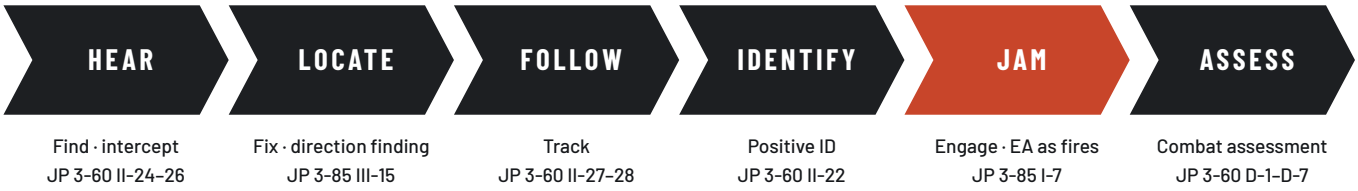
Every contact is a track: an estimate with a covariance, not ground truth. Bearings update it through an unscented Kalman filter (Julier sigma points, $\kappa = 0$, circular mean for angles), which matches filterpy to 3×10^{-12}. Prediction uses central-difference state transition matrices and white-noise acceleration (Bar-Shalom et al. 2001). Two cases needed special handling.

- **Close in.** With HYDRA a few kilometres from the inspector, sigma points can straddle the sensor. Within a tenth of the range, a bearing therefore enters as a pseudo-linear measurement (Lindgren & Gong 1978; Aidala & Nardone 1982). Between craft in nearly the same orbit, range is unobservable from angles alone without a manoeuvre (Woffinden & Geller 2009). So without a recent radar range, the track keeps at least 0.5 × range of 1σ along the line of sight (design value). In 42 looks from alongside to departure, no estimate fell outside 3σ.
- **Unseen burns.** A bearing whose normalised innovation passes the $\chi^2(1)$ 99.9% point (10.83) is taken as a burn nobody saw, and widens the track (Bar-Shalom ch. 11). The 50 m/s and 10 min used are design values.

A track nobody has detected for 24 h, or whose 1σ exceeds 100,000 km, is dropped (after Blackman & Popoli 1999). Both thresholds are design values. Contacts are drawn after MIL-STD-2525D, with a dashed frame for a last known position.

8 DOCTRINE AND THE KILL CHAIN

FIG. 12 THE GAME'S VERBS AS A KILL CHAIN



Each step mapped to its doctrinal term and citation (table below).

The game's verbs follow US joint doctrine, read from the published texts: JP 3-85 *Joint Electromagnetic Spectrum Operations* (2020)[139], JP 3-60 *Joint Targeting* (2018)[141], AFDP 3-60 (2026)[142] and the USSF *Space Warfighting* framework (2025)[143].

GAME STEP	DOCTRINAL TERM	CITATION
Hear	F2T2EA *Find*; electromagnetic support search and intercept	JP 3-60 II-24–26; JP 3-85 I-6
Locate	*Fix*; direction finding, precision geolocation	JP 3-60 II-26–27; JP 3-85 III-15, B-4
Follow	*Track*	JP 3-60 II-27–28
Identify	Positive identification by EW support systems or visual recognition (doctrine acquires it during *Fix*)	JP 3-60 II-22
Jam	*Engage*; electromagnetic attack as a form of fires	JP 3-85 I-7, I-13, GL-6, GL-7
Assess	Combat assessment, with confidence graded confirmed, probable or possible	JP 3-60 D-1–D-7

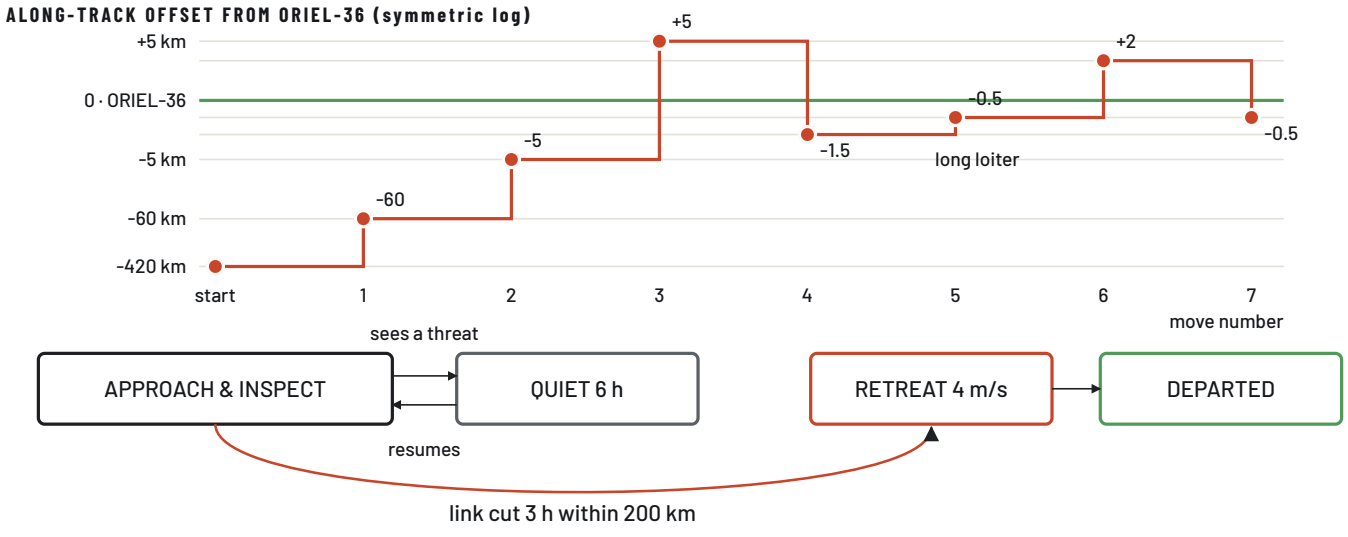
Under JP 3-85, a stealth ship going dark is practising electromagnetic protection: "multispectral low-observability" is one of its listed examples. Doctrine says nothing about TDOA/FDOA, look-through, own-platform jammer isolation, or an escalation ladder. JP 3-85, JP 3-60 and AFDP 3-60 never use the words "spoofing" or "reversible": their terms are intrusion and deception, and temporary or nonlethal effects. (The USSF framework does speak of reversible and nonreversible actions, pp. 5 and 9.) The escalation meter is therefore entirely a design value. Its ordering (reversible and deniable acts cheap, debris-generating acts ruinous) follows doctrine's escalation-of-force guidance qualitatively.

9 THE SCENARIO AND ITS REAL-WORLD BASIS

THE SHADOWING is modelled on a real uncooperative rendezvous and proximity operation against a commercial X-band radar-imaging satellite in October 2026. It was reported from LeoLabs tracking data by Breaking Defense and The War Zone on 19 October 2026[158,159]. The events are moved to Jupiter and the craft are fictional. As reported, the real sequence was: coplanar alignment; closing from long range; station-keeping within 5 km, then 1–2 km; a closest pass reported at 500 m (another report gives 26 m); moves ahead of and behind the target; and a controlled departure into a larger orbit. The inspector's seven moves follow that sequence: −60, −5, +5, −1.5, −0.5 (the long loiter), +2 and −0.5 km along-track.

- **Orbit.** 1.3 R_J (92,940 km), a 4.37 h period. This is inside Jupiter's radiation belts, so every craft is assumed rad-hardened (Juno's titanium vault cuts its dose about 800 times[161]). No dose is modelled.
- **The inspector's radio.** An X-band (8.436 GHz) BPSK burst at 256 kbaud, 10 W into a 0.4 m dish aimed at its handler, 90 s long, after each move. These are design values within ordinary CCSDS practice. Seeing a threat, it goes quiet for 6 hours.
- **Losing its link.** Cut off from its handler for 3 hours within 200 km of its target, and with a Blue ship within 25 km of it (a design value: one MATCH drifts about 20 km in 3 h), it backs away prograde at 4 m/s and departs. Cut off but alone, it holds its slot and waits. Taking its controls needs the jammer holding its owner off first. This is grounded in practice, not in a published rule: a Progress approach to the ISS was aborted on loss of its telemetry link in 2010[162], and ESA's ATV carried RETREAT and ESCAPE modes for the same reason. The threshold and the burn are DESIGN VALUES.
- **Timescale.** The real event took weeks. The game compresses it to about 57 hours over three days DESIGN VALUE.

FIG. 13 HOW THE INSPECTOR BEHAVES



Top: the seven along-track station points from §9, on a symmetric log scale so the last few hundred metres are visible. Bottom: what makes it go quiet or leave. Thresholds are DESIGN VALUES.

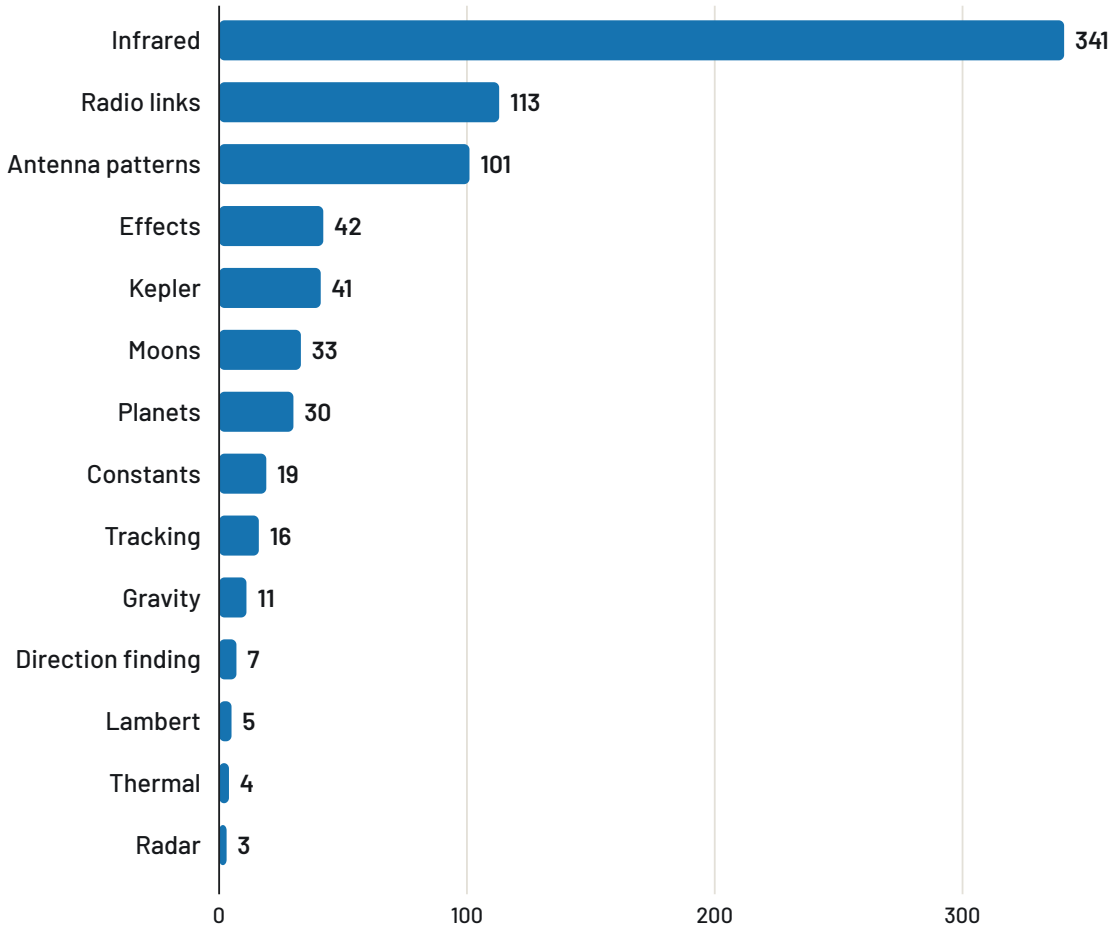
10 VERIFICATION

A Godot script runs every test case through the game's own functions and writes the results out. A Python suite (pytest) then compares them with independent references, each comparison carrying its own tolerance. JPL data is cached, so the suite runs offline. On 25 October 2026 it made 762 comparisons across 63 tests; all 762 fall within tolerance, and 4 further rows report known simplifications.

MODEL	REFERENCE	TOLERANCE	WORST RESULT
Kepler propagation (122 cases, e to 0.999 and hyperbolic)	hapsira farnocchia/vallado; DOP853	1e-8	2.3e-10
Lambert (129 transfers)	hapsira izzo	1e-6	1.9e-12
Worked examples	Curtis Ex. 3.7, 5.2; Vallado Ex. 2-4	½ last digit	pass
Force model (J2, third bodies)	hapsira perturbations; gradient of the J2 potential	1e-9	9e-11
RK4 integration, 1 day	DOP853, rtol 1e-13	5 m	0.03 m
J2 secular rates	Vallado eqs. 9-41/42	1%	0.40%
Moons and planets	JPL Horizons	per body	Table 1
Antenna patterns (468 cases)	ITU-R SA.509-3, S.465-5, coded from the text	1e-9 dB	1.4e-14
Free-space loss	ITU-R P.525-4 eq. (4), 32.4 dB constant	0.05 dB	0.048 dB
Radiometer gain √(Bτ)	Monte Carlo energy detector, 2,000 trials	6%	1.5%
Interferometer DF σ	Monte Carlo two-element correlator	8%	3%
Radar SNR and monopulse angle	Skolnik; Barton	1e-9; 0.2%	6e-15; 0.14%
Jammer J/S and reach	Adamy, independent dB form; brentq	1e-9 dB	3e-14
In-band Planck share	astropy BlackBody, 60–3,000 K	1e-6	6e-11
Radiators against JIMO	Siamidis 2005 design point	2%	1.1%
Shadow fraction	Dense numpy disc integration	0.02	0.019
Skin heat capacity	scipy solve_ivp	—	0.0002 K
Unscented update	filterpy UKF	1e-6	3e-12
Rocket equation	Tsiolkovsky, every ship	0.006 km/s	0.0045 km/s

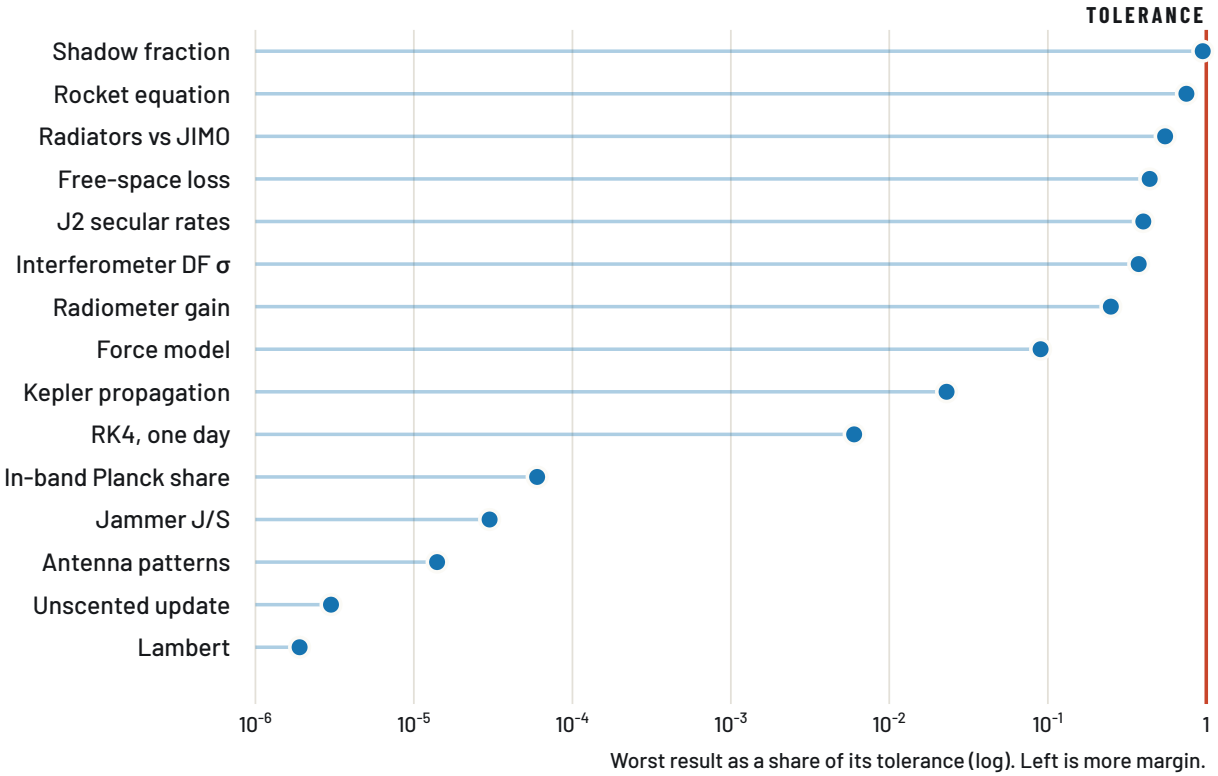
Table 5. Selected comparisons. The full table has one row per comparison.

FIG. 14 WHERE THE 766 ROWS GO



Rows per domain on 25 October 2026: 762 within tolerance, and 4 that report known simplifications.

FIG. 15 **HOW MUCH MARGIN EACH CHECK HAS**



Worst result divided by its tolerance, for the numeric rows of Table 5. Every dot sits left of the red line; most sit orders of magnitude inside it.

The suite has also found and fixed real errors. Moon longitudes were up to 114° off. DF σ was three times too large, because it was fed the integrated SNR. The barrage band was twice as wide as the real allocation. Long-way Lambert transfers failed. Kepler's equation lost accuracy near e = 1. Each fix and its effect on play is recorded in the project's verification log.

11 LIMITS, SIMPLIFICATIONS AND OPEN QUESTIONS

Not public, so extrapolated

- Jammer, uplink and handler EIRPs. Counter Communications System and Meadowlands power is not published.
- Weapon-class laser wall-plug efficiency. The game's 25% is conservative against industrial fibre lasers.
- Time-transfer accuracy between platforms at Jupiter, where there is no GPS.

Not verified

- Mid-infrared zodiacal background at 5.2 AU, and Jupiter's band-resolved brightness (a grey body at 125.6 K is used).
- HawkEye 360 geolocation accuracy; the Albersheim validity limits; specific-emitter identification accuracy; look-through duty cycles.
- The 2024 edition of JP 3-60 (the 2018 edition was read), and JP 5-0's definitions of MOE and MOP.
- MIL-STD-2525D was read through a reproduction of its tables, not the original PDF.

Simplifications

- Moons are planar mean-element ellipses with no mutual perturbations (≤ 23,000 km and ≤ 0.72°).
- Radiators lie flat to the plane, and the effect of jamming is binary (link cut or not). Doctrine describes a continuum from temporary to permanent and from partial to complete.
- The umpire's view: some kill-chain indicators still read ground truth rather than Blue's assessed belief.
- Red doesn't yet adapt by frequency hopping, nulling or switching to optical links. All of these are real (ITU-R SM.1055; Cummings 1992; NASA DSOC).
- No radiation dose is modelled.

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Every source the game cites, in its code, data and tests, grouped by the model that uses it. A link opens the document itself, or a free copy where the publisher's is behind a paywall. Works with no readable copy online are listed without one.

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